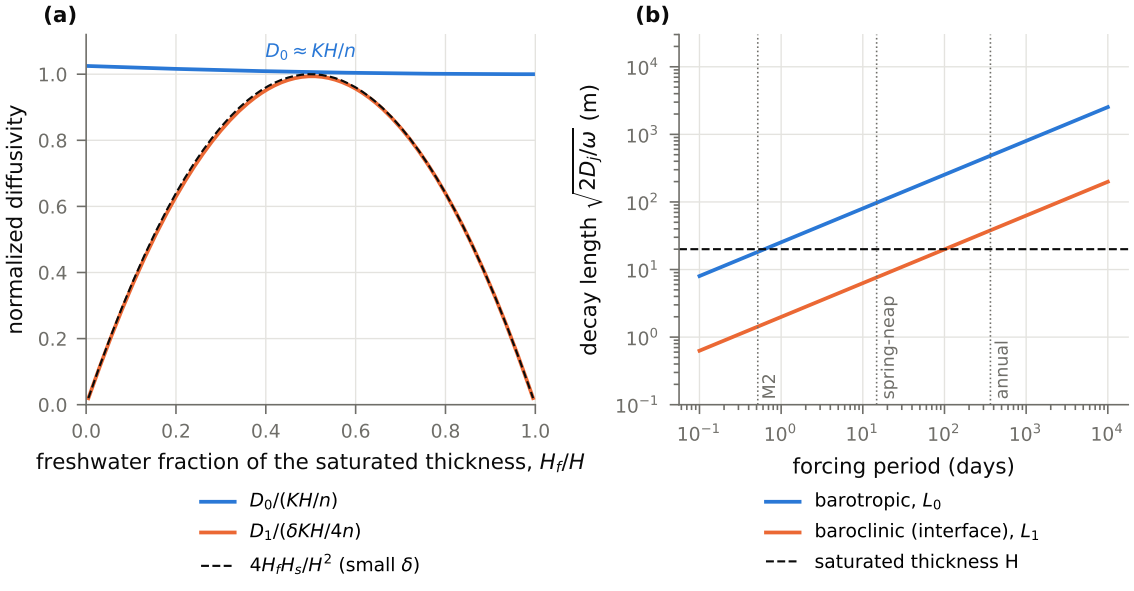
(a)
$D_0 \approx KH/n$
normalized diffusivity
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
freshwater fraction of the saturated thickness, H_f/H
$D_0/(KH/n)$
$D_1/(\delta KH/4n)$
$4H_fH_s/H^2$ (small δ)
(b)
decay length $\sqrt{2D_J/\omega}$ (m)
10^4
10^3
10^2
10^1
10^0
10^{-1}
M2
spring-neap
annual
10^{-1}
10^0
10^1
10^2
10^3
10^4
forcing period (days)
barotropic, L_0
baroclinic (interface), L_1
saturated thickness H